02/05

Wednesday, 5 February 2025 9:30 AM

FD / FV

Finite Differences / finite element

approximates derivatives at nodes in a grid

control volume analysis (dividing up domain in control volume & integrating equations over these volumes)

Infinite no. of possible FD/FV eqs for given PDE

U_i^n → exact soln of PDE at (x_i, t^n)

u_i^n → exact soln of FDE/FVE at (x_i, t^n) ← no round off errors

N_i^n = numerical soln of FDE / FVE

Error at (x_i, t^n)

$$E_i^n = U_i^n - N_i^n = \underbrace{(U_i^n - u_i^n)}_{\text{discretization error}} + \underbrace{(u_i^n - N_i^n)}_{\text{round off / Stability crit}}$$

Consistency vs. Convergence vs. Stability

"identical twins"

(looks the same but does not mean convergence or stable) → Have to just do Taylor series expansion

Discretizing a PDE: → Han

If there is a ratio $\left(\frac{\Delta t}{\Delta x}\right)$ → then it will be conditionally stable

Stability Analysis

+ Discrete perturbation → Put a disturbance in soln. at some time & see its growth

+ Fourier (von Neumann) → disturbance at every grid point

+ Matrix

+ Modified eqn.

Fourier: $\varepsilon(x,t) = \frac{a_0}{2} + \sum (a_j \cos k_j + b_j \sin k_j x)$ $\quad k_j = \frac{2\pi j}{L}$

$\cos\psi = \frac{(e^{i\psi} + e^{-i\psi})}{2}$; $\sin\psi = \frac{(e^{i\psi} - e^{-i\psi})}{2i}$

Fourier; $f(x,t) = \frac{a_0}{2} + \sum (a_j \cos k_j x + b_j \sin k_j x)$ $\quad k_j = \frac{2\pi j}{L}$

+ Modified eqn.

$\cos\psi = \frac{(e^{i\psi} + e^{-i\psi})}{2}$; $\sin\psi = \frac{(e^{I\psi} - e^{-I\psi})}{2I}$

↳ But this must also be discretized
(there is an associated error for this as well)

→ Unconditional vs. conditional

$$\varepsilon_i = \sum_{j=-N}^{N} A_j^n \exp(I k_j x_i) \qquad \varepsilon_{i,t}^{n+1} = \sum A_j^{n+1} \exp$$

+ As you refine Δx & ΔT

$\frac{\Delta t}{\Delta x}$ } conditional (when discretized)